

GO-CAMPUS

DYNAMIC CAMPUS SHUTTLE TRACKING AND LOAD MONITORING SYSTEM

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PROBLEM STATEMENT

- Students currently have no access to real-time shuttle location or arrival updates, causing uncertainty about shuttle schedules. As a result, students often spend excessive time waiting at shuttle stops without knowing when the shuttle will arrive, leading to frustration and wasted time especially during peak academic hours.
- The campus shuttle system does not provide real-time information on shuttle occupancy, meaning students only discover that a shuttle is full after it arrives. This results in overcrowding, discomfort, and students being forced to wait for the next shuttle, causing delays and affecting punctuality for classes or appointments.
- Due to the lack of automatic data collection on shuttle usage such as real-time passenger count and shuttle movement administrators are unable to optimize shuttle routes or scheduling efficiently. This leads to poor resource allocation, inefficient route planning, and inconsistent shuttle frequency, reducing overall service effectiveness.

PROJECT OBJECTIVES

The main objective of this project is to create an IoT-based system that improves the efficiency of campus shuttle services. The system will show real-time shuttle location and passenger load so students can avoid long waiting times and overcrowded shuttles. By combining sensors, GPS tracking, and a cloud dashboard, the system helps students plan their travel better and provides useful data for campus management to improve shuttle schedules and routes.

- To study the current campus shuttle operations and identify issues related to waiting time, overcrowding, and lack of real-time information.
- To develop an IoT-based system that integrates GPS tracking and passenger counting sensors, and transmits real-time shuttle location and occupancy data to a cloud platform accessible through a mobile or web dashboard.
- To evaluate the system's accuracy, performance, and usability by testing real-time data transmission, dashboard responsiveness, and effectiveness in reducing waiting time and overcrowding.

DEVELOPMENT PROGRESS

No	Activities	Expected Completion Date	Actual Completion Date	Progress Status
1.0	System Architecture & Cloud Foundation		-	
1.1	Firebase Core, Auth, & Realtime DB Integration	2026-03-01	2026-03-01	100% Completed
1.2	Dual-Sync Security Rules (App vs ESP32)	2026-03-03	2026-03-04	100% Completed
2.0	Identity & Access Management (IAM)			
2.1	Multi-Role Registration (Student & Driver)	2026-03-05	2026-03-06	100% Completed
2.2	Profile Config (Full UniSZA Faculty/Course Dataset)	2026-03-07	2026-03-07	100% Completed
2.3	Role-Based Terminal Entry (Admin/Student/Driver)	2026-03-10	2026-03-11	100% Completed
3.0	IoT Hardware & Telemetry Linkage			

DEVELOPMENT PROGRESS

No	Activities	Expected Completion Date	Actual Completion Date	Progress Status
3.1	ESP32 IR Sensor Integration (Passenger Counting)	2026-03-12	2026-03-13	100% Completed
3.2	Live GPS Telemetry Sync (Bus Node Location)	2026-03-15	2026-03-16	100% Completed
3.3	Bi-Directional Reset Handshake (App-to-ESP32)	2026-03-17	2026-03-17	100% Completed
4.0	Driver Command Center			
4.1	Mission Control Dashboard (Live Broadcast Toggle)	2026-03-18	2026-03-19	100% Completed
4.2	Real-time Load Monitor (Circular Node Visuals)	2026-03-20	2026-03-21	100% Completed
4.3	Automated Shift History (Local Session Logs)	2026-03-22	2026-03-23	100% Completed
5.0	Student HUD & Interactive Tracker			

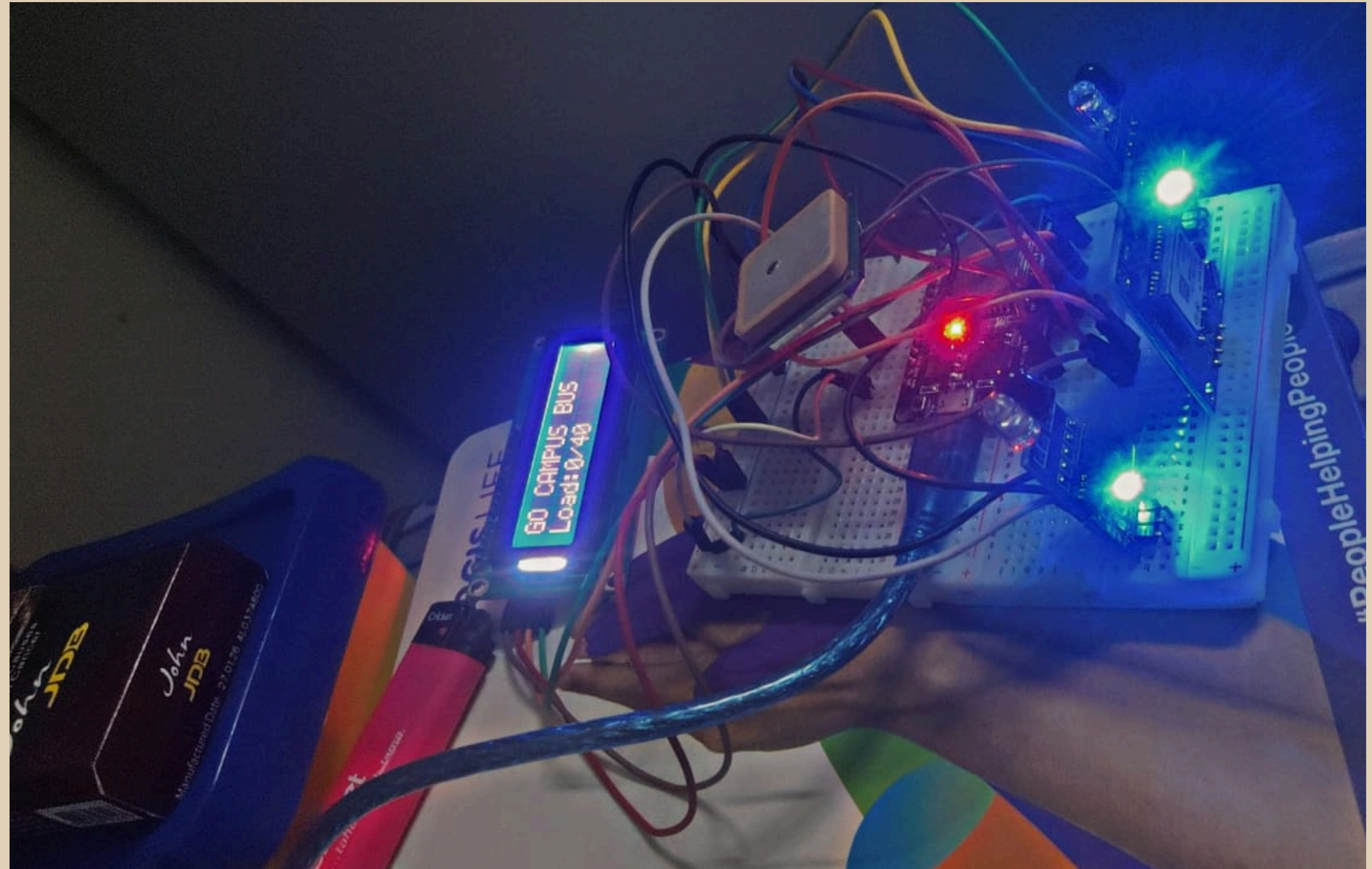
DEVELOPMENT PROGRESS

No	Activities	Expected Completion Date	Actual Completion Date	Progress Status
5.1	Radar Tracking System (Distance & ETA Logic)	2026-03-24	2026-03-25	100% Completed
5.2	Nearest Bus Targeter (Camera Snap Logic)	2026-03-26	2026-03-26	100% Completed
5.3	Student Alert System (Node Signal Bell/Buzzer)	2026-03-28	2026-03-28	100% Completed
6.0	Admin Archives & Reporting			
6.1	System Archives (Historical Date-Query Engine)	2026-03-30	2026-03-31	100% Completed
6.2	Automated PDF Report Export (Daily Transit Logs)	2026-04-02	2026-04-02	100% Completed
7.0	Quality Assurance & Deployment	2026-04-05	2026-04-03	95% Finalizing
7.1	UI/UX Refinement (Holographic HUD Aesthetic)			

DEVELOPMENT PROGRESS

No	Activities	Expected Completion Date	Actual Completion Date	Progress Status
7.2	Final Stress Testing (Software-Hardware Sync)	2026-04-10	-	60% In Progress
7.3	Nearest Bus Targeter (Camera Snap Logic)	2026-03-26	-	Pending

PROJECT DEMO



THANK YOU

